COMPUTER NETWORKS

Top 50 Interview Questions & Answers

Ultimate Preparation Guide (Hinglish Edition)

Yeh booklet aapko engineering and IT interviews me computer networking ke concepts clear karne me madad karegi. Isme core fundamentals se lekar troubleshooting aur security tak ke sabhi topics aasan bhasha me samjhaye gaye hain.

Designed for: Software Engineers, Network Engineers, and System Administrators

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1. OSI & TCP/IP Models

Q1. OSI Model ki saari 7 layers ke naam aur unke main functions kya hain?

OSI (Open Systems Interconnection) model me 7 layers hoti hain:

1. **Physical Layer (Layer 1):** Bits ko physical media (cables, fiber, radio) ke zariye transmit karti hai.
2. **Data Link Layer (Layer 2):** Physical addressing (MAC address) handle karti hai aur error-free frame delivery karti hai. (Devices: Switch, Bridge).
3. **Network Layer (Layer 3):** Logical addressing (IP address) manage karti hai aur packets ko source se destination tak route karti hai. (Device: Router).
4. **Transport Layer (Layer 4):** End-to-end communication, reliability (TCP/UDP), flow control, aur segmentation provide karti hai.
5. **Session Layer (Layer 5):** Multi-devices ke beech connection session ko establish, manage, aur terminate karti hai.
6. **Presentation Layer (Layer 6):** Data format transfer (syntax), encryption/decryption, aur data compression karti hai.
7. **Application Layer (Layer 7):** User directly isse interact karta hai. (Protocols: HTTP, FTP, SMTP, DNS).

Q2. TCP/IP Model aur OSI Model mein kya mukhyah (main) antar hai?

- OSI ek theoretical reference model hai (7 layers), jabki TCP/IP ek practical model hai (4 layers: Network Access, Internet, Transport, Application) jise internet me real-world use kiya jata hai.
- OSI model protocol-independent hota hai, jabki TCP/IP model protocols par hi customized hai.

Q3. Router, Switch aur Hub OSI model ki kis-kis layer par kaam karte hain?

- **Hub:** Physical Layer (Layer 1) par kaam karta hai. Yeh sirf signals ko sabhi ports par repeat (broadcast) karta hai bina smart decision liye.
- **Switch:** Data Link Layer (Layer 2) par kaam karta hai. Yeh MAC addresses ke basis par dedicated ports par data frames bhejta hai.
- **Router:** Network Layer (Layer 3) par kaam karta hai. Yeh IP addresses ke basis par packets ko path select (route) karke alag-alag networks me bhejta hai.

Q4. Data Encapsulation aur Decapsulation kya hota hai?

- **Encapsulation:** Jab data Application layer se Physical layer ki taraf niche aata hai, toh har layer apna header (control information) usme add karti hai. Ise Encapsulation kehte hain.
- **Decapsulation:** Receiver side par jab data physical layer se upar application layer tak jata hai, toh har layer apna respective header nikal (strip-off) deti hai. Ise Decapsulation kehte hain.

Q5. Layer 2 aur Layer 3 switches mein kya farq hai?

- **Layer 2 Switch:** Sirf MAC addresses read kar sakta hai. Yeh sirf ek hi LAN segment ke andar devices ko connect karta hai aur routing nahi kar sakta.
- **Layer 3 Switch:** MAC address ke sath IP address bhi read kar sakta hai. Yeh switching ke sath-sath VLANs ke beech internal routing (Inter-VLAN routing) bhi kar sakta hai.

Q6. PDU (Protocol Data Unit) kya hota hai? Bits, Frames, Packets aur Segments kis layer par hote hain?

PDU data ka representation hai jo har layer par alag naam se jana jata hai:

- **Segments:** Transport Layer (Layer 4)
- **Packets:** Network Layer (Layer 3)
- **Frames:** Data Link Layer (Layer 2)
- **Bits:** Physical Layer (Layer 1)

Q7. Application layer protocols (HTTP, FTP, SMTP) ke default port numbers kya hain?

Aamtaur par pooche jaane wale ports:

- HTTP: Port 80 | HTTPS: Port 443
- FTP: Port 20, 21 | SMTP: Port 25
- DNS: Port 53 | DHCP: Port 67, 68
- SSH: Port 22 | Telnet: Port 23

2. IP Addressing & Subnetting

Q8. IPv4 aur IPv6 mein kya major differences hain?

Feature	IPv4	IPv6
Size	32-bit address	128-bit address
Format	Dotted Decimal (e.g., 192.168.1.1)	Hexadecimal Colon (e.g., 2001:db8::1)
Total IPs	Approx. 4.3 Billion (~2^{32})	Practically Infinite (~2^{128})
Security	IPSec optional hai	IPSec built-in (mandated) hai

Q9. MAC Address (Physical) aur IP Address (Logical) mein kya farq hai?

- **MAC Address:** Yeh 48-bit ka unique hardware ID hota hai jo manufacturing ke samay network interface card (NIC) par permanently burn kiya jata hai. Yeh local network delivery ke liye use hota hai.
- **IP Address:** Yeh 32-bit (IPv4) ya 128-bit (IPv6) ka logical address hota hai jo network administrator ya DHCP assign karta hai. Yeh global packet routing ke liye use hota hai.

Q10. Subnetting kya hoti hai aur iski zaroorat kyun padti hai?

Ek bada physical IP network ko do ya do se zyada chote, logical subnetworks me divide karne ki process ko **Subnetting** kehte hain.
Zaroorat: IPs ki wastage rokne ke liye, network congestion aur broadcast domain ko chota karne ke liye, aur network security behtar karne ke liye.

Q11. Classful aur Classless IP addressing (CIDR) mein kya antar hai?

- **Classful Addressing:** Isme IP addresses pre-defined classes (Class A, B, C, D, E) me segmented hote hain jinka subnet mask fix hota hai (Class A: /8, B: /16, C: /24). Isme IP ka nuksaan bahut hota hai.
- **Classless Addressing (CIDR - Classless Inter-Domain Routing):** Isme fixed class boundary nahi hoti. Subnet mask ki length flexible hoti hai aur slash notation (jaise 192.168.1.0/26) se likhi jaati hai.

Q12. Private IP aur Public IP addresses mein kya farq hai?

- **Private IP:** Local LAN networks ke andar use hota hai (unrouted on internet). E.g., 192.168.x.x. Yeh free hote hain.
- **Public IP:** Internet se communicate karne ke liye zaroori hote hain. Inhe IANA/ISP manage karta hai aur yeh paid/unique hote hain.

Q13. Loopback address kya hota hai aur iska use kab kiya jata hai?

IPv4 me **127.0.0.1** aur IPv6 me **::1** loopback address hote hain. Iska use local system ka TCP/IP stack self-test karne ke liye kiya jata hai. Agar aap `ping 127.0.0.1` karte hain aur reply aata hai, iska matlab aapke computer ka network stack sahi chal raha hai.

Q14. Subnet Mask kya batata hai? Network ID aur Host ID kaise nikaalte hain?

Subnet Mask batata hai ki IP address ka kaunsa hissa **Network ID** hai aur kaunsa **Host ID**.
E.g., IP: `192.168.1.50`, Mask: `255.255.255.0` (matlab /24). Binary logical AND operation karne par pehle 24 bits 1s ho jaate hain, isliye Network Portion: `192.168.1.0` aur Host Portion: `.50` hai.

Q15. NAT (Network Address Translation) kya hai aur yeh kaise kaam karta hai?

NAT ek process hai jiske zariye multiple devices ke Private IPs ko router par ek single Public IP me translate kiya jata hai taaki local LAN devices internet ko access kar sakein. Isse Public IP addresses ki kaafi bachat hoti hai.

3. Transport Layer & Protocols (TCP vs UDP)

Q16. TCP aur UDP mein kya antar hai? Dono ke real-life use cases bataiye.

- **TCP (Transmission Control Protocol):** Connection-oriented hai. Har packet ka acknowledgement hota hai. Agar data loss ho jaye toh use retransmit karta hai. Use cases: Web browsing, Email, File transfer.
- **UDP (User Datagram Protocol):** Connectionless hai. Koi reliability aur acknowledgement nahi deta, isliye fast hota hai. Use cases: Video streaming, Online gaming, DNS.

Q17. TCP Three-Way Handshake process kya hai? Yeh connection kaise banata hai?

TCP data transfer shuru karne se pehle 3 step me connection banata hai:

1. **SYN (Synchronize):** Sender ek packet bhejta hai jisme SYN flag set hota hai (apna initial sequence number batane ke liye).
2. **SYN-ACK:** Receiver acknowledge karta hai aur apna sequence number + response SYN-ACK packet bhejta hai.
3. **ACK:** Sender confirm karne ke liye final ACK packet bhejta hai. Iske baad connection establish ho jata hai.

Q18. TCP mein Flow Control aur Congestion Control kaise manage hota hai?

- **Flow Control:** Receiver ke buffer capabilities ke hisab se data rate limit karna, jo **Sliding Window** protocol se hota hai.
- **Congestion Control:** Network path me congestion (bheed) dekh kar transmission rate kam karna, jisme algorithms jaise **Slow Start, Congestion Avoidance, Fast Retransmit** use hote hain.

Q19. Windowing (Sliding Window Protocol) kya hota hai?

Yeh ek mechanism hai jahan sender ek baar me kitna data frames (Window Size) bhej sakta hai bina acknowledgement ke, use decide kiya jata hai. Isse network bandwidth efficiently use hoti hai aur flow control maintain rehta hai.

Q20. Agar TCP reliable hai, toh hum real-time video call mein UDP kyun use karte hain?

Video call me speed aur continuity sabse important hoti hai. Agar network me koi frame kharab ho ya der se aaye, toh TCP use re-request karega, jisse call lag (atakegi/freeze) karegi. UDP me bina wait kiye naye frames continuous aate rehte hain jisse live feel bana rehta hai, bhale hi thodi picture quality dropped ho jaye.

Q21. TCP Connection Termination (Four-Way Handshake) kaise hota hai?

TCP connection band karne ke liye 4 steps leta hai:

1. Active system bhejta hai **FIN** packet.
2. Saamne wala system bhejta hai **ACK**. (Half-closed state)
3. Kuch der baad dusra system apna bacha kaam khatam karke bhejta hai **FIN**.
4. Pehla system bhejta hai final **ACK**, aur tab connection puri tarah se close ho jata hai.

4. Application Layer Protocols & Web

Q22. DNS (Domain Name System) kya hai? Yeh IP address ko domain name mein kaise badalta hai?

DNS internet ki "Phone Book" hai. Yeh human-readable domain names (jaise google.com) ko machine-readable IP addresses (jaise 142.250.190.46) me translate karta hai. Jab aap search karte hain, DNS hierarchy (Root server, TLD server, Authoritative server) ke zariye answer search karke device ko batata hai.

Q23. DNS Lookup process mein Recursive aur Iterative queries mein kya farq hai?

- **Recursive Query:** Aapka client local DNS resolver se full resolution chahta hai. Resolver jab tak IP dhoodh nahi leta ya error nahi deta tab tak puri mehnat khud karta hai.
- **Iterative Query:** Ek DNS server dusre DNS server ko poochta hai. Agar use answer pata hai toh de dega, warna kisi aur server ka referral (address) de dega jo target ke close ho.

Q24. HTTP aur HTTPS mein kya antar hai? SSL/TLS handshake kya hota hai?

- **HTTP:** Plain text format me data transfer karta hai. Yeh secure nahi hai (unencrypted, Port 80).
- **HTTPS:** SSL/TLS certificate ka use karke traffic ko encrypted banata hai (secure, Port 443).
- **SSL/TLS Handshake:** Jab client-server connect hote hain, toh dono asymmetric cryptography ke zariye secure session key negotiate karte hain, certificate verify karte hain taaki data safe rahe.

Q25. DHCP (Dynamic Host Configuration Protocol) kaise kaam karta hai? (DORA Process kya hai?)

DHCP devices ko automatically IP addresses assign karta hai. Isme **DORA** process hoti hai:

- **Discover (D):** Client pure network me broadcast karta hai IP maangne ke liye.
- **Offer (O):** DHCP servers client ko available IP offer karte hain.
- **Request (R):** Client selected IP ko lock karne ki request bhejta hai.
- **Acknowledgment (A):** DHCP server confirm karke IP lease detail client ko de deta hai.

Q26. FTP (File Transfer Protocol) aur TFTP (Trivial FTP) mein kya difference hai?

- **FTP:** Robust, reliable (TCP based, Ports 20, 21), secure, user authentication support karta hai. Large files transfer ke liye standard hai.
- **TFTP:** Simple, lightweight (UDP based, Port 69), bina password aur fast transfer ke liye use hota hai (jaise network boot files).

Q27. SMTP, IMAP aur POP3 protocols mein kya farq hai (Email networking)?

- **SMTP:** Mail send karne/outgoing ke liye use hota hai.
- **POP3:** Server se local device par mail download kar leta hai aur server se mail delete ho jata hai (Single-device focus).
- **IMAP:** Mails ko server par hi manage karta hai aur client sirf sync karta hai (Multi-device access support karta hai).

Q28. ARP (Address Resolution Protocol) aur RARP mein kya antar hai?

- **ARP:** Jab kisi device ko samne wale ka IP pata ho par uska hardware MAC address na pata ho, toh ARP use MAC pata lagane me help karta hai.
- **RARP (Reverse ARP):** MAC address ke basis par IP address pata lagane ke liye use hota tha (ab iska kam DHCP karta hai).

5. Routing & Switching Concepts

Q29. Routing aur Switching mein kya basic difference hai?

- **Switching:** LAN ke andar local devices ke beech frames transfer karta hai using MAC Addresses. (Same network).
- **Routing:** Do ya do se zyada alag-alag logical IP networks ke beech data packets forward karta hai using IP Addresses. (Different networks).

Q30. Unicast, Multicast, aur Broadcast traffic mein kya farq hai?

- **Unicast:** One-to-One communication. Ek specific computer se dusre specific computer tak data bhejna.
- **Multicast:** One-to-Many (Selected). Ek system se selective group of devices tak data bhejna.
- **Broadcast:** One-to-All. Ek network me present sabhi systems ko ek sath data bhej dena.

Q31. Broadcast Storm kya hota hai aur ise kaise roka jata hai?

Jab do switches ke beech multiple active redundant path hote hain aur frames gol-gol ghumte rehte hain (switching loops), toh excessive broadcast packets pura bandwidth consume kar lete hain. Isse network jam ho jata hai jise Broadcast Storm kehte hain. Isko **STP (Spanning Tree Protocol)** chala kar roka jata hai.

Q32. VLAN (Virtual LAN) kya hai aur iska use kyun kiya jata hai?

VLAN ek high-performance physical switch ko logically alag-alag isolated virtual networks me divide karne ki process hai. Isse security badhti hai aur logical teams (jaise HR, IT, Finance) ka broadcast domain alag ho jata hai bhale hi wo ek hi hardware switch se connected hon.

Q33. Spanning Tree Protocol (STP) kya hai? Loops ko kaise rokta hai?

STP (IEEE 802.1D) ek layer 2 loop-avoidance protocol hai. Yeh switch networks me redundant links detect karta hai aur backup links ko "blocking state" me daal deta hai, loop nahi hone deta. Jab main link break hota hai, toh backup link automatically active kar deta hai.

Q34. Routing Table kya hoti hai aur isme kaun-kaun si details hoti hain?

Router ke RAM me save ek list hoti hai jisse router ko path selection karne me direction milti hai. Isme:

- Destination Network ID
- Subnet Mask
- Next-Hop (Agla router ya Gateway ka IP)
- Metric / Cost (Route kitna accha hai)
- Interface (Kaunse port se packet bhejhein)

Q35. Static Routing aur Dynamic Routing mein kya farq hai?

- **Static Routing:** Admin khud manually router table me path configure karta hai. Small networks ke liye secure aur zero CPU overhead deta hai.
- **Dynamic Routing:** Routers automatic routing protocols (OSPF, BGP) ke zariye khud path sikhate hain aur path failure hone par alternate route choose karte hain. Large networks ke liye suitable hai.

Q36. IGP (Interior Gateway Protocol) aur EGP (Exterior Gateway Protocol) mein kya antar hai?

- **IGP:** Ek hi Autonomous System (Organization/Enterprise) ke andar use hone wale routing protocols. E.g., OSPF, EIGRP, RIP.
- **EGP:** Alag-alag autonomous systems ke beech routing karne ke liye design protocols. E.g., BGP (Border Gateway Protocol) jo pure Internet ko chalata hai.

Q37. Distance Vector (RIP) aur Link State (OSPF) routing protocols mein kya farq hai?

- **Distance Vector (RIP):** Routing decision hop counts (routers in path) ke upar hoti hai. Yeh pure routing table apne neighbors ko periodic intervals par bhejte hain.
- **Link State (OSPF):** Routing decision link status, bandwidth aur cost par hoti hai using Dijkstra's shortest-path algorithm. Yeh sirf network changes hone par incremental updates bhejte hain. Fast convergence hoti hai.

Q38. BGP (Border Gateway Protocol) kya hai aur yeh kahan use hota hai?

BGP ek Path-Vector protocol hai jise internet ka "Backbone protocol" kaha jata hai. Yeh baday networks aur Autonomous Systems (AS) ke beech path information share karta hai taaki optimal internet routing establish ho sake.

6. Network Security & Advanced Concepts

Q39. Firewall kya hoti hai? Packet Filtering aur Stateful Inspection firewall mein kya farq hai?

Firewall ek network security device hai jo predefined rules ke mutabiq incoming/outgoing traffic monitor aur control karti hai.

- **Packet Filtering (Stateless):** Sirf packet ke individual headers (Source/Dest IP, Port) check karti hai bina session context jaane.
- **Stateful Inspection:** Poore active connection/session ke state ko monitor karti hai aur valid replies ko hi aane deti hai.

Q40. VPN (Virtual Private Network) kya hai aur yeh data ko kaise secure karta hai?

VPN internet ke unstable public network ke upar ek encrypted, secure virtual tunnel banata hai jo remote user ko company LAN network se safe connect karta hai. Yeh **IPSec** ya **SSL/TLS** protocols use karke data ko encrypt karta hai taaki bich me koi data read na kar sake.

Q41. Symmetric aur Asymmetric encryption mein kya antar hai?

- **Symmetric Encryption:** Data encrypt aur decrypt karne ke liye ek hi Single secret key use ki jaati hai (Fast, useful for bulk data transfer). E.g., AES.
- **Asymmetric Encryption:** Isme do keys (Public key - encryption ke liye, Private key - decryption ke liye) use hoti hain. Highly secure par thodi computational heavy hoti hai. E.g., RSA.

Q42. Proxy Server kya hota hai aur yeh Reverse Proxy se kaise alag hai?

- **Forward Proxy:** Client side par lagta hai. Internal users ko internet hide/restrict karke safe access dilwata hai.
- **Reverse Proxy:** Server side par lagta hai. Client ko nahi pata hota ki asal server kaunsa hai. Yeh load balancing, caching aur SSL termination ke liye kam aata hai.

Q43. DoS (Denial of Service) aur DDoS attacks kya hote hain?

- **DoS:** Ek target machine/server ko dher saari fake requests bhej kar overwhelm kar dena taaki regular users ke liye service crash ho jaye. Source single system hota hai.
- **DDoS:** Distributed DoS attack me hazaaron 'zombie bots' (compromised devices) se ek sath target server par hamla kiya jata hai, jise identify/block karna bahut mushkil hota hai.

Q44. Spoofing aur Phishing mein kya difference hai?

- **Spoofing:** Kisi malicious action ke liye apni identity badalna (e.g., IP address badal ke target ko lagna ki packet trustable source se aa raha hai).
- **Phishing:** Ek duplicate, trusted dikhne wala page/email design karke user ko fasaana taaki wo apna password, credit card info khud de baithe.

Q45. IDS (Intrusion Detection System) aur IPS (Intrusion Prevention System) mein kya farq hai?

- **IDS:** Passive system hai. Yeh network traffic scan karta hai aur agar koi suspect activity dikhe toh admin ko alert bhejta hai (Detection focus).
- **IPS:** Active system hai. Yeh threat detect hone par alert bhejne ke sath-sath active step le kar use block bhi kar deta hai (Prevention focus).

7. Troubleshooting & Practical Scenarios

Q46. Ping aur Traceroute commands ke piche kaun sa protocol kaam karta hai? Dono mein kya farq hai?

Dono **ICMP (Internet Control Message Protocol)** ka use karti hain.

- **Ping:** Check karta hai host active hai ya nahi, aur delay (RTT) batata hai (Echo Request/Reply).
- **Traceroute:** IP header ke **TTL (Time to Live)** parameter ko step-by-step badha kar yeh trace karti hai ki source se destination ke beech kitne hops (routers) hain aur kaun-kaun se bad route par hain.

Q47. Agar aap kisi website ko ping kar paa rahe hain par browser mein open nahi ho rahi, toh kya problem ho sakti hai?

- Website ka web server (HTTP service/Port 80 ya Port 443) down ho sakta hai bhale hi uska basic ICMP stack live ho.
- Aapka firewall outgoing HTTP/HTTPS traffic block kar raha ho.
- Browser proxy ya caching settings corrupt ho sakti hain.

Q48. Default Gateway kya hota hai? Agar yeh galat configure ho toh kya hoga?

Default Gateway aapke local switch/router ka LAN IP hota hai jo local network ko bahar (Internet ya doosre networks) se jodta hai.
Galat Configuration: Agar yeh galat ho, toh aap apne LAN me sabhi devices se sahi connect ho payenge, par external websites ya networks (jaise Internet) par data nahi bhej payenge.

Q49. ipconfig (Windows) ya ifconfig/ip a (Linux) command ka kya use hai?

In commands ka use local system ke network card interface details check karne ke liye hota hai. Isme system ka current IP, Subnet Mask, Default Gateway, aur MAC address detail show hoti hai.

Q50. Jab aap browser mein 'https://www.google.com' type karke Enter press karte hain, toh step-by-step kya hota hai?

1. **URL parsing:** Browser URL parse karta hai aur 'HTTPS' check karta hai.
2. **DNS Lookup:** Local cache, host file, router, ISP resolver, aur external DNS systems se hotey hue `google.com` ka IP target hota hai.
3. **TCP Handshake:** Browser target IP ke port 443 par connect hota hai (SYN -> SYN-ACK -> ACK).
4. **TLS/SSL Handshake:** Secure cryptographic keys exchange hoti hain.
5. **HTTP GET Request:** Browser web page ka default HTML code maangta hai.
6. **Response Processing:** Google web server page return karta hai, browser HTML render karta hai aur static resources (images, JS files) download karke display par dikhata hai.